BODDU SESHA VENKATA SAI RANGA AKHILESH

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EDUCATION

B. Tech in Electronics and Communication Engineering with Specialisation in Artificial Intelligence & Machine Learning (2021 – 2025)

Deenbandhu Chhotu Ram University of Science and Technology, Murthal

CGPA: 8.32

PROFESSIONAL EXPERIENCE

Project Scientist, Centre for Atmospheric Sciences (CAS), Indian Institute of Technology Delhi

July 2025 - July 2026

- Architected and built a full-stack AI evidence-intelligence platform for collecting, preserving, searching and analysing governance and policy documents from webpages, PDFs, DOCX files, scanned records and structured uploads.
- Designed a service-oriented architecture integrating React, TypeScript, Express, FastAPI, PostgreSQL, pgvector, Redis and Docker based development and production environments.
- Built scalable BullMQ and Celery workers for ingestion, OCR, embedding, tagging and bulk processing, incorporating idempotent execution, bounded concurrency, telemetry, retries and failure recovery.
- Engineered page-complete extraction using browser rendering, native document parsers, OCRmyPDF and Tesseract while retaining page-level processing status, character offsets and inspectable evidence locators.
- Implemented hybrid retrieval using PostgreSQL full-text search, 1,536-dimensional pgvector embeddings, HNSW indexing, provenance-aware chunking, cosine-similarity filtering and optional LLM reranking.
- Developed evidence-constrained RAG, citation validation and grounding controls that verified quotations, detected unsupported claims, removed invalid citations and generated evidence-gap reports.
- Established audit-ready provenance linking source captures, SHA-256 hashes, revisions, extraction settings, chunks, embeddings, tags, model runs, citations and generated claims.
- Designed and evaluated an audio-ML pipeline for source apportionment in horn-dominated traffic recordings, combining audio-SLM synchronisation, spectral feature engineering, unsupervised clustering and transformer-based horn classification.
- Investigated engine, tyre, exhaust and horn signatures using GMM, NMF and AST models, documenting limitations caused by overlapping sources, recording variability and limited labelled data.

INTERNSHIP

Research Intern in Electronics and Data Analysis using Machine Learning, Indian Institute of Technology Delhi

January 2025 - June 2025

- Engineered and implemented multi-sensor (CO_2 & $PM_{2.5}$, Temperature, Humidity, Gyroscope, Accelerometer and Geomagnetic) IoT systems by designing circuits, achieving real-time data logging, remote monitoring via web server integration and battery-powered operation.
- Developed a dedicated air-velocity-transducer data logger, mapping power and signal connections and addressing microcontroller signal fluctuations.
- Automated calibration for 17 logger nodes against three LICOR reference datasets using Python, incorporating timestamp harmonisation, one-minute averaging, outlier filtering, linear regression and sensor-error analysis against reference measurements.
- Modeled and fabricated sensor enclosures, battery holders, airflow reducers and acrylic calibration chambers using Fusion 360, 3D printing and laser cutting; conducted airflow checks to refine diffuser placement.
- Assembled and tested multiple CO_2 and PM logger prototypes, prepared PM logger units for IIT Jammu and delivered a two-day workshop on data logger construction to IIT Delhi students.

Electronics Trainee, Centre for Development of Telematics, New Delhi

July 2023 - August 2023

- Programmed analog and digital blocks on PSoC microcontrollers and contributed to two LCD-control schematic designs.
- Prototyped application-specific peripherals utilising hardware description languages.
- Programmed an LCD controller that reduced UI load time by 20% and supported scrolling text and image rendering.

SKILLS

Programming Languages: Python, SQL, JavaScript, TypeScript, Embedded C/C++, HTML, CSS

AI, Machine Learning & Data Science: Artificial Intelligence, Machine Learning, Deep Learning, Data Analysis, Data Analytics, Statistics, Predictive Analytics, scikit-learn, Keras, NumPy, Pandas, Matplotlib, Plotly

Generative AI, NLP & Retrieval: Large Language Models, OpenAI API, Retrieval-Augmented Generation (RAG), Natural Language Processing, embeddings, semantic search, hybrid retrieval, sentence-transformers, spaCy, KeyBERT, YAKE, pgvector, HNSW indexing, LLM reranking, citation grounding, provenance tracking, OCRmyPDF, Tesseract, document parsing, browser-based extraction

Audio Machine Learning & DSP: Audio preprocessing, STFT, log-mel spectrograms, MFCCs, spectral feature engineering, librosa, SciPy, windowing, segmentation, HTS-AT, Audio Spectrogram Transformer (AST)

Backend & Web Development: React, Node.js, Express.js, FastAPI, REST APIs, Prisma, Puppeteer, web extraction, API security

Databases & Infrastructure: PostgreSQL, Redis, Celery, BullMQ, Docker, Docker Compose

Data Governance: Content fingerprinting, SHA-256 integrity validation, revision tracking, audit trails

Embedded Systems & Electronics: Internet of Things, rapid prototyping, PSoC 5LP (ARM-based microcontroller), Raspberry Pi, Arduino (ATmega328P), Seeed Studio XIAO ESP32-S3, I²C, SPI

Prototyping Tools: Autodesk Fusion 360, 3D printing and laser cutting

PROJECTS

Density-Based Intelligent Traffic Control System
August 2024 - December 2024

- Integrated Raspberry Pi 4, YOLOv10, OpenCV, and Python into an AI-based traffic signal control system to estimate vehicle density and dynamically adjust signal timing.
- Built a real-time object detection and vehicle counting pipeline with a Tkinter-based GUI for traffic density visualisation.
- Explored edge-computing feasibility for deploying computer vision models on low-power hardware for smart-city applications.

Smart Cane for Visually Impaired People - An IIT Roorkee Sponsored Project
May 2023 - September 2023

- Redesigned a smart cane to detect overhead obstacles, ground-level objects, potholes and stairs using multiple sensors.
- Integrated four vibration motors, five LEDs and a buzzer to provide location-specific obstacle alerts.
- Produced a 3D-printed prototype to reduce component complexity and improve manufacturing feasibility.

VOLUNTEERING

Campaign Volunteer - Prototype Mentoring, Change Makers 2025 Summer Bootcamp, IIT Delhi
June 2025

- Mentored student teams in converting early-stage concepts into practical prototype designs by evaluating functionality, component placement, and build feasibility.
- Guided the integration of sensors, electronic components and control systems into physical prototypes while addressing compatibility and implementation challenges.

CERTIFICATIONS

- Advanced Aerosol Science Intensive Course - CAS, IIT Delhi & INAR, University of Helsinki
- Introduction to Internet of Things; Deep Learning; Deep Learning for Natural Language Processing - IIT Kharagpur
- Computer Vision and Image Processing - IIT Guwahati

ACHIEVEMENTS

- Secured **2nd Place** out of 30 participants in Tinker and Build in Technova competition conducted by DCRUST Technical Fest
- Received a **₹1 lakh fellowship** at the State-Level Hackathon organised by DCRUST and IIT Roorkee, ranking among the top solutions from 40+ teams.
- Qualified **GATE** EC 2024 and GATE DA 2025